

CO3-AT2-LABORATORY PERFORMANCE

Course: DSA0103-Object Oriented Programming with C++

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1. Write a c++ program to create a class for a student with a constructor and a destructor

Code:

```
1  #include <iostream>
2  using namespace std;
3  class Student
4  {
5  private:
6      int rollNo;
7      string name;
8  public:
9      Student(int r, string n)
10     {
11         rollNo = r;
12         name = n;
13         cout << "Constructor called" << endl;
14     }
```

```
15     void display()
16     {
17         cout << "\nStudent Details" << endl;
18         cout << "Roll No: " << rollNo << endl;
19         cout << "Name: " << name << endl;
20     }
21     Student()
22     {
23         cout << "Destructor called" << endl;
24     }
25 };
```

```
26 int main()
27 {
28     int roll;
29     string name;
30     cout << "Enter Roll Number: ";
31     cin >> roll;
32     cin.ignore();
33     cout << "Enter Student Name: ";
34     getline(cin, name);
35     Student s(roll, name);
36     s.display();
37     return 0;
38 }
```

INPUT

101
riya

Output:

```
Enter Roll Number: Enter Student Name: Constructor called

Student Details
Roll No: 101
Name: riya
Destructor called
```

2. Write a c++ program to create a class for a book with a constructor and a destructor

Code:

```
1 #include <iostream>
2 using namespace std;
3 class Book
4 {
5 private:
6     int bookId;
7     string title;
8 public:
9     Book(int id, string t)
10    {
11        bookId = id;
12        title = t;
13        cout << "Constructor called" << endl;
14    }
15
16    void display()
17    {
18        cout << "\nBook Details" << endl;
19        cout << "Book ID: " << bookId << endl;
20        cout << "Book Title: " << title << endl;
21    }
22    Book()
23    {
24        cout << "Destructor called" << endl;
25    }
26 };
```

```

26 int main()
27 {
28     int id;
29     string title;
30     cout << "Enter Book ID: ";
31     cin >> id;
32     cin.ignore();
33     cout << "Enter Book Title: ";
34     getline(cin, title);
35     Book b(id, title);
36     b.display();
37     return 0;
38 }

```

INPUT

```

201
C++ Programming

```

Output:

```

Enter Book ID: Enter Book Title: Constructor called

Book Details
Book ID: 201
Book Title: C++ Programming
Destructor called

```

3. Write a c++ program to create a class for a rectangle with a constructor and a destructor

Code:

```

1 #include <iostream>
2 using namespace std;
3 class Rectangle
4 {
5 private:
6     int length, width;
7 public:
8     Rectangle(int l, int w)
9     {
10         length = l;
11         width = w;
12         cout << "Constructor called" << endl;
13     }

```

```

14     void display()
15     {
16         cout << "\nRectangle Details" << endl;
17         cout << "Length: " << length << endl;
18         cout << "Width: " << width << endl;
19         cout << "Area: " << length * width << endl;
20     }
21     Rectangle()
22     {
23         cout << "Destructor called" << endl;
24     }
25 };

```

```

26 int main()
27 {
28     int length, width;
29     cout << "Enter Length: ";
30     cin >> length;
31     cout << "Enter Width: ";
32     cin >> width;
33     Rectangle r(length, width);
34     r.display();
35     return 0;
36 }

```

INPUT

10
5

Output:

```

Enter Length: Enter Width: Constructor called

Rectangle Details
Length: 10
Width: 5
Area: 50
Destructor called

```

4. Write a c++ program to create a class for a car with a constructor and a destructor

Code:

```
1 #include <iostream>
2 using namespace std;
3 class Car
4 {
5 private:
6     string brand, model;
7 public:
8     Car(string b, string m)
9     {
10         brand = b;
11         model = m;
12         cout << "Constructor called" << endl;
13     }
```

```
14 void display()
15 {
16     cout << "\nCar Details" << endl;
17     cout << "Brand: " << brand << endl;
18     cout << "Model: " << model << endl;
19 }
20 Car()
21 {
22     cout << "Destructor called" << endl;
23 }
24 };
```

```
25 int main()
26 {
27     string brand, model;
28     cout << "Enter Car Brand: ";
29     getline(cin, brand);
30     cout << "Enter Car Model: ";
31     getline(cin, model);
32     Car c(brand, model);
33     c.display();
34     return 0;
35 }
```

INPUT

Toyota
Innova

Output:

```
Enter Car Brand: Enter Car Model: Constructor called

Car Details
Brand: Toyota
Model: Innova
Destructor called
```

5. Write a c++ program to create a class for a bank account with a constructor and a destructor

```
1 #include <iostream>
2 using namespace std;
3
4 class BankAccount
5 {
6 private:
7     int accountNumber;
8     string accountHolder;
9     float balance;
10 public:
11     BankAccount(int accNo, string name, float bal)
12     {
13         accountNumber = accNo;
14         accountHolder = name;
15         balance = bal;
16         cout << "Constructor called" << endl;
17     }
18
19     void display()
20     {
21         cout << "\nBank Account Details" << endl;
22         cout << "Account Number: " << accountNumber << endl;
23         cout << "Account Holder: " << accountHolder << endl;
24         cout << "Balance: " << balance << endl;
25     }
26     BankAccount()
27     {
28         cout << "Destructor called" << endl;
29     }
30 };
31
```

```

30 int main()
31 {
32     int accNo;
33     string name;
34     float balance;
35
36     cout << "Enter Account Number: ";
37     cin >> accNo;
38     cin.ignore();
39     cout << "Enter Account Holder Name: ";
40     getline(cin, name);
41     cout << "Enter Balance: ";
42     cin >> balance;
43     BankAccount b(accNo, name, balance);
44     b.display();
45     return 0;
46 }

```

INPUT

```

2005
santhosh
10000

```

Output:

```

Enter Account Number: Enter Account Holder Name: Enter Balance:
Constructor called

Bank Account Details
Account Number: 2005
Account Holder: santhosh
Balance: 10000

```